

AROOJ ABBAS

BSCS Student · Backend & AI Systems Developer

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PROFESSIONAL SUMMARY

Final-year Computer Science student (Punjab University, Lahore) building ARIA, a multi-agent agentic AI desktop assistant powered by a LangGraph-based tool router. Own the backend architecture for four agent modules — Gmail Email Agent, WhatsApp Messaging Agent, Document Summarizer, and Browser Automation. Combine hands-on experience in agentic AI frameworks (LangChain, LangGraph, Groq) with a solid foundation in full-stack development (React.js, Node.js, Python, MongoDB). Seeking an internship or entry-level role in AI/ML engineering or backend development.

TECHNICAL SKILLS

AI / Agentic Systems: LangChain, LangGraph (multi-agent tool routing), Groq API, ChromaDB (vector storage), Whisper (speech-to-text), Retrieval-Augmented Generation (RAG) fundamentals

Backend & APIs: Python, Node.js, Flask, Gmail API, REST APIs, Playwright (browser automation), SQL, MongoDB

Frontend: React.js, JavaScript (ES6+), HTML5, CSS3, Tailwind CSS

Tools & Platforms: Git, GitHub, VS Code, Postman, Jira, Figma, PyInstaller

FEATURED PROJECT

ARIA — AI Research & Intelligent Assistant (Final Year Project, 4-person team) | 01/2026 – Present

A multi-agent, agentic AI desktop assistant built around a LangGraph-based tool router that dynamically dispatches user requests to specialized agents. Own Backend 1: architecture and implementation of four core agents.

- Designing and building the Gmail Email Agent for automated email reading, drafting, and triage via the Gmail API.
- Developing a WhatsApp Messaging Agent for automated message handling and response generation.
- Building a Document Summarizer that condenses long-form documents into structured summaries.
- Implementing Browser Automation workflows using Playwright to let the assistant complete web-based tasks autonomously.
- Collaborating with a 4-person team (AI/ML lead, backend, frontend) on system architecture, scoping, and a 6-month delivery roadmap.

OTHER PROJECTS

Restaurant Management System — React.js, Node.js, JavaScript, HTML, CSS | 12/2025 – Present

- Building a full-stack platform to streamline communication between restaurant management and clients, with role-based views for orders and updates.

Personal Finance Tracker — JavaScript, HTML5, CSS3

- Built an interactive finance management app with expense tracking and Chart.js data visualization, using local storage for persistence.

Health Tracking Application — JavaScript, HTML5, CSS3

- Built a calorie counter with meal planning and nutrition tracking, using a responsive CSS Grid/Flexbox layout.

PROFESSIONAL EXPERIENCE

Teaching Assistant — Computer Organization & Assembly Language | Punjab University, Lahore | 02/2025 – 07/2025

- Supported 50+ students in understanding low-level computing concepts and Assembly language programming.
- Led weekly lab sessions on MIPS/ARM architecture, memory hierarchy, and I/O systems.
- Graded assignments and gave feedback on assembly code optimization; authored supplementary practice materials.

Computer Science Teacher | Inspire Education School System, Muridke | 01/2024 – 02/2025

- Taught computer science and mathematics to secondary students; developed lesson plans and programming exercises.

EDUCATION

BS Computer Science | *University of Punjab, Lahore (PU)* | 2023 – 2027
Final-year student; current focus area: agentic AI systems and multi-agent architectures (Final Year Project: ARIA).
F.Sc. Pre-Engineering | *Govt. Girls College, Muridke* | 2021 – 2022 | 94%

CERTIFICATIONS & ACHIEVEMENTS

- Web Design & Development Certification — PNY Trainings, Arfa Tower (01/2026)
- Frontend Development Certification — freeCodeCamp
- Python Programming Certificate — Punjab University SDC (2024)
- Python Programming Internship — CosmiCode (2025)
- Certification in Graphic Designing — LWMG Academy (03/2025)
- PEEF Scholarship, Lahore Board (2023); Quaid-e-Azam Scholarship, Lahore Board (2022)
- Participant — Electron'24 (UET Lahore), CosmiCode Programming Competition (2024), COSMOCON'25 (Punjab University)